

HANDS ON – Keerthivasan R DAY-30

Problem Statement: Implement Dependency Injection in ASP.NET Core

Background

A growing e-commerce company wants to build a modular and maintainable web application using **ASP.NET Core MVC**. The development team wants to reduce tight coupling between components and improve testability of the application.

To achieve this, they decide to implement **Dependency Injection (DI)** using the built-in IoC container of **ASP.NET Core**.

Objectivej

Develop a simple Product Management module that demonstrates the implementation of **Dependency Injection** in ASP.NET Core.

Functional Requirements

The system should:

1. Display a list of products.
2. Add a new product.
3. Retrieve product details by ID.
4. Delete a product.

Technical Requirements

1 Create an Interface

- Define an interface **IProductService** with methods:
 - GetAllProducts()
 - GetProductById(int id)
 - AddProduct(Product product)
 - DeleteProduct(int id)

Create a Service Class

- Implement the interface in `ProductService`.
- Add sample in-memory data (no database required for simplicity).

Register Services

- Register the service in `Program.cs` using:
 - `AddScoped()`
 - `AddTransient()`
 - or `AddSingleton()`

Inject Dependency

- Inject `IProductService` into:
 - `ProductController` via Constructor Injection.

Key Learning Outcomes

Students should understand:

- What is Dependency Injection?
- Why DI reduces tight coupling.
- Difference between:
 - Transient Lifetime
 - Scoped Lifetime
 - Singleton Lifetime
- Constructor Injection in ASP.NET Core.
- How DI improves unit testing.

Expected Output

- Application runs successfully.
- Controller accesses service without creating object manually.
- No usage of `new ProductService()` inside controller.
- Proper separation of concerns followed.

Evaluation Criteria

- Proper interface implementation.
- Correct service registration.
- Clean architecture.
- Code reusability.
- Understanding of DI lifetimes.

CODE:

Model/ Product.cs

```
namespace ProductManagementSystem.Models
{
    public class Product
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public double Price { get; set; }
    }
}
```

Service/ IProductService.cs

```
using ProductManagementSystem.Models;

namespace ProductManagementSystem.Services
{
    public interface IProductService
    {
        IEnumerable<Product> GetAllProducts();

        Product GetProductById(int id);

        void AddProduct(Product product);

        void DeleteProduct(int id);
    }
}
```

Service/ ProductService.cs

```
using ProductManagementSystem.Models;

namespace ProductManagementSystem.Services
{
    public class ProductService : IProductService
    {
        private static List<Product> _products = new()
        {
            new Product { Id = 1, Name = "Laptop", Price = 1200 }
        };
        public IEnumerable<Product> GetAllProducts() => _products;

        public Product? GetProductById(int id) => _products.FirstOrDefault(p => p.Id == id);

        public void AddProduct(Product product) => _products.Add(product);

        public void DeleteProduct(int id) => _products.RemoveAll(p => p.Id == id);
    }
}
```

Controller/ ProductController.cs

```
using Microsoft.AspNetCore.Mvc;
using ProductManagementSystem.Models;
```

```

using ProductManagementSystem.Services;

namespace ProductManagementSystem.Controllers
{
    public class ProductController : Controller
    {
        private readonly IProductService _productService;

        public ProductController(IProductService productService)
        {
            _productService = productService;
        }

        // 1. Display List
        public IActionResult Index()
        {
            var products = _productService.GetAllProducts();
            return View(products);
        }

        // 3. Retrieve Details by ID
        public IActionResult Details(int id)
        {
            var product = _productService.GetProductById(id);
            if (product == null) return NotFound();
            return View(product);
        }

        // 2. Add Product (The Form Page)
        public IActionResult Create() => View();

        [HttpPost]
        public IActionResult Create(Product product)
        {
            _productService.AddProduct(product);
            return RedirectToAction("Index");
        }

        // 4. Delete Product
        public IActionResult Delete(int id)
        {
            _productService.DeleteProduct(id);
            return RedirectToAction("Index");
        }
    }
}

```

Program.cs

```

using ProductManagementSystem.Services; // Add this using

var builder = WebApplication.CreateBuilder(args);

builder.Services.AddScoped<IProductService, ProductService>();

// Add services to the container.
builder.Services.AddControllersWithViews();

var app = builder.Build();

// Configure the HTTP request pipeline.
if (!app.Environment.IsDevelopment())

```

```

{
    app.UseExceptionHandler("/Home/Error");
    // The default HSTS value is 30 days. You may want to change this for production
    scenarios, see https://aka.ms/aspnetcore-hsts.
    app.UseHsts();
}

app.UseHttpsRedirection();
app.UseStaticFiles();

app.UseRouting();

app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");

app.Run();

```

OUTPUT:

```

Info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:7199
Info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5005
Info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
Info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
Info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\keerthi\OneDrive\Desktop\upGrad\DAY-31\ProductManagementSystem\ProductManagementSystem

```

ProductManagementSystem
Home
Privacy

No data to display.

Product List

Add New Product

Name	Price	Actions
laptop	₹ 50,000.00	Details Delete
Clock	₹ 1,000.00	Details Delete
TV	₹ 30,000.00	Details Delete

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Product Details

ID: 0

Name: laptop

Price: ₹ 50,000.00

[Back to List](#)

Add New Product

Product Name

Air Purifier

Price

10000

Save Product